A structural label in the subgroup table of A_{11}

human_compression project

Connor and Leemans' companion table for A_{11}, dated 27 November 2012, prints $2 \cdot A_5 : 2$ in row 186, of order 240 and class length 20790 [2]. The native page is page 11. Section 3.2 of their article distinguishes the nonsplit middle dot from the general extension symbol and explicitly adopts left association [1, p. 263]. Thus the printed expression denotes $(2 \cdot A_5) : 2$ under the stated convention. We prove that this interpretation is impossible in A_{11}. We then identify the actual subgroup supplied by the independently checked numerical table. The original expression remains in the source ledger; the corrected structure is recorded separately.

Lemma 1. *A nonsplit extension of A_5 by a normal subgroup of order two is perfect.*

Proof. Write $1 \to Z \to N \xrightarrow{q} A_5 \to 1$, with $|Z| = 2$. The subgroup Z is central, since its automorphism group is trivial. Recall that A_5 is perfect. For completeness, the commutator of $(12)(34)$ and $(12)(35)$ is a 3-cycle. The commutator subgroup therefore contains every 3-cycle, since these form one conjugacy class in A_5; they generate A_5. Consequently $q(N') = A_5$. If $N' \cap Z = 1$, then N' is a complement to Z, contradicting the nonsplitting assumption. Hence $Z \subseteq N'$ and $N' = N$. □

Proposition 2. *No nonsplit extension $2 \cdot A_5$ embeds in A_{11}. In particular, no subgroup of A_{11} has the structure $(2 \cdot A_5) : 2$.*

Proof. Suppose that such a subgroup N exists, and let z be its central involution. Since z is an even permutation on eleven points, it is a product of $m = 2$ or $m = 4$ disjoint transpositions. Put $\Omega = \mathrm{supp}(z)$. Centrality makes Ω invariant under N, and restriction defines a homomorphism

$$N \longrightarrow C_{\mathrm{Sym}(\Omega)}(z) \cong C_2^m \rtimes S_m.$$

Indeed, a permutation commuting with z permutes its two-point orbits and independently interchanges the two points in each orbit. The group on the right is solvable: its base group is abelian, and S_2 and S_4 are solvable. One may use the normal series $1 \triangleleft V_4 \triangleleft A_4 \triangleleft S_4$ to see the latter. The image of the perfect group N is perfect, while every perfect solvable group is trivial. But z has nontrivial restriction to Ω, a contradiction. The final assertion follows because the indicated semidirect product contains $2 \cdot A_5$ as a subgroup. □

This obstruction concerns every subgroup of A_{11}; it does not depend on the choice of a correspondence between the numerical source rows and the computed representatives. It does not assert that the authors intended the impossible interpretation. A missing pair of parentheses would change the expression to a valid one.

Proposition 3. *The saved representative corresponding to row 186 is*

$$H = \langle (12345),\ (12)(8\,10\,9\,11),\ (8\,9)(10\,11) \rangle.$$

It is isomorphic to

$$\{(\sigma, k) \in S_5 \times C_4 : \mathrm{sgn}(\sigma) = (-1)^k\}.$$

Its structure can be written $(A_5 \times 2) \cdot 2$, or equivalently $2 \cdot (A_5 : 2)$, with the displayed parentheses.

Proof. Put $r = (8\,10\,9\,11)$, acting on a set disjoint from the first five points. Each displayed generator belongs to the indicated pullback. The first two generators project onto S_5: a 5-cycle and an adjacent transposition generate S_5. Moreover, the square of the second generator is $r^2 \neq 1$. The projection kernel therefore has order at least two. The pullback has order $2|S_5| = 240$, so containment and projection show that H is the entire pullback.

The elements with even k form $N = A_5 \times \langle r^2 \rangle$, of index two. Every element outside N has odd k, so its square has second coordinate two and cannot be the identity. Thus the extension of C_2 by N has no complement and is nonsplit. This gives the first expression. Projection onto S_5 has central kernel $\langle r^2 \rangle$. Every lift of a transposition has square r^2; a section of this projection would preserve the order of a transposition, which is impossible. Hence this central extension is nonsplit as well. Finally, $S_5 = A_5 : C_2$, with the usual nontrivial conjugation action, giving the second expression. □

The accompanying literal certificate uses only permutations on eleven points. It closes the saved generators and checks equality with all 240 pullback elements. It verifies the direct product of order 120, the absence of involutions outside it, and all twenty lifts of the ten transpositions. As an additional check, it propagates all eight possible assignments of the three generators to C_2 across every Cayley edge. Exactly one assignment is surjective, and its kernel is the displayed $A_5 \times C_2$. The centre of that kernel has order two. Thus the index-two subgroup cannot itself be a nonsplit extension $2 \cdot A_5$.

The table and its naming convention are due to Connor and Leemans. The arguments above use elementary facts about central extensions, perfect groups, and permutation centralizers. The contribution here is the explicit correction and its connection to the checked representative; no general priority claim is made for those facts. This note corrects one structural entry. It neither changes the numerical subgroup lattice nor proves an open conjecture.

References

[1] T. Connor and D. Leemans, *An atlas of subgroup lattices of finite almost simple groups*, Ars Mathematica Contemporanea **8** (2015), 259–266.

[2] T. Connor and D. Leemans, *The subgroup lattice of* A_{11}, companion table, 27 November 2012, 41 pages; row 186 on page 11.